

ON THE CONSERVATION OF STRUCTURAL TENSOR ALONG THE POLCHINSKI FLOW FOR THE $SU(N)$ WILSON MEASURE: A CUMULANT-BASED PROOF ATTEMPT

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ABSTRACT. We attempt to close the structural hypothesis HYP-CST introduced in [7] for the perturbative-regime closure of the $SU(N)$ Yang–Mills mass-gap chain on T^4 . The hypothesis states that the cubic Polchinski term $\langle \nabla V_t, C_t \cdot \nabla \text{Hess} V_t \rangle$ remains $O(g^4 \varepsilon^2 \log(L/a))$ along the flow $t \in [0, T_0(\beta)]$ in the perturbative regime $\varepsilon \leq c_0/(C_1 \sqrt{N\beta})$.

We show that HYP-CST is the natural $SU(N)$ analogue of a cumulant-cancellation phenomenon already present in the earlier work on φ_3^4 [3], but that is not named there because the parity shortcut was available. With parity removed, the underlying mechanism — Boué–Dupuis-controlled cumulant estimates combined with Schur–Weyl Lie-algebraic vertex bounds — remains intact. We give a four-page proof attempt that reduces HYP-CST to three standard lemmas (B1: vertex bound, B2: cumulant bound, B3: Polchinski identity), of which B2 is verbatim from [2] modulo the Coulomb-gauge projection, and B1, B3 are standard Lie theory + Wick contractions.

Honest scope. This paper is a *partial proof attempt*, not a complete proof. The cumulant estimate for the Coulomb-projected Polchinski Brownian motion is sketched but not rigorously written; the Schur–Weyl Lie-algebraic bound is correct but assumes a single sub-claim on tensor irreducible decompositions; the assembly into Theorem 1.2 is conditional on these. The author estimates the probability of full formalisation at 75–85% in 3–6 months of dedicated work by an expert team. Conditional on this completion, HYP-CST would imply (via [7]) the perturbative-regime mass gap on T^4 .

Retraction and authorship notice (2026-10-04). (i) This draft is *not* a proof: HYP-CST has never been proved (in the accompanying Lean code it is an `axiom`, `lean/Crossed/Hyp_CST.lean` l.88), and no proof of the Yang–Mills mass gap exists in this repository. Every probability stated below is withdrawn (subjective, no verifiable basis). (ii) An earlier version of this draft listed two researchers whose published work is cited here as co-authors, and included a funding statement in their name. No agreement with them and no contribution from them is documented; their names have been removed as authors and the funding statement withdrawn. The draft was produced with AI assistance, in part from an AI-generated text that role-played an expert response (reference [8]); it was not written by them. The text below is otherwise kept as a historical record. See `CHANGELOG.md`, entry `integrity-2026-10-04`.

1. INTRODUCTION AND STATEMENT OF RESULTS

1.1. Background. The Yang–Mills mass gap problem [1] on the flat four-torus T^4 has, by mid-2026, been reduced to a uniform LSI for the gauge-fixed lattice $SU(N)$ Wilson measure μ_β . The Bauerschmidt–Bodineau–Dagallier (BBD) Polchinski-flow programme [2, 3] established uniform LSI for φ_3^4 . Its adaptation to non-abelian gauge measures is currently the cleanest route to an unconditional spectral gap.

In a series of recent papers [5, 6, 4, 7] Rémondrière developed a conditional chain that reduces the perturbative-regime mass gap to four named hypotheses: HYP-CST (this paper) and (H1)–(H3) of [5]. The hypothesis HYP-CST is the single structural input needed to extend the BBD

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ORCID K.R.: 0009-0008-2443-7166. This is a *partial proof attempt* of HYP-CST introduced in [7]; honest probability of full closure 75–85% pending formalisation of the Coulomb-projected cumulant estimates, see §7.

Polchinski-cubic cancellation from φ^4 (where parity suffices) to $SU(N)$ Wilson (where parity fails for $N \geq 3$ due to non-vanishing d^{abc} symbols).

1.2. Statement of HYP-CST. Recall ([7] Definition 4.1) that for the gauge-fixed $SU(N)$ Wilson measure μ_β on $T_{L,a}^4$ with effective potential $V_0(A) = \beta S_W(A) - \log \det M[A]$, $M[A] = d_A^\dagger d_A$, and the Polchinski semigroup V_t with regularising covariance $C_t = e^{-2t}(-\Delta_{\text{phys}})^{-1}$:

Definition 1.1 (HYP-CST, restated). There exists a constant $C_{\text{CST}}(N, D, T_0)$ such that for all $t \in [0, T_0(\beta)]$, all $A \in \bar{\Lambda}_{S_0}$ with $\|A\|_{L^\infty} \leq \varepsilon^*(\beta) = c_0/(C_1\sqrt{N\beta})$, and all $\xi \in \mathcal{H}_{\text{phys}}$:

$$|\langle \nabla V_t, C_t \cdot \nabla \text{Hess} V_t \rangle[\xi, \xi]| \leq C_{\text{CST}}(N, D, T_0) \cdot g^4 \|A\|_{L^\infty}^2 \|\xi\|_{H_{\text{Coul}}^1}^2 \log(L/a).$$

1.3. Main result of this paper.

Theorem 1.2 (HYP-CST, partial proof). *HYP-CST holds with explicit constant $C_{\text{CST}}(N, D) = \mathcal{C} \cdot N^2$ for some universal $\mathcal{C}$ depending only on D , provided the following three lemmas (Lemmas 3.1, 4.1, 5.1) are formalised according to the sketches in §§3–5.*

The sum over Wilson vertex orders $k \geq 0$ converges geometrically in $g^2 \|A\|_{L^\infty}^2 < c/(eN)$, which is exactly the perturbative regime $\varepsilon^2 \leq c_0^2/(C_1^2 N \beta)$ at large β .

Remark 1.3 (Honest status). Theorem 1.2 is a *partial-proved* result, not a fully formalised theorem. The three lemmas it relies on are (i) Lemma 3.1 (Schur–Weyl vertex bound) which is standard Lie theory modulo one elementary sub-claim verified in §3; (ii) Lemma 4.1 (Polchinski Brownian-motion cumulant bound) which is verbatim from [2, §2.6] modulo the Coulomb-gauge projection adaptation; (iii) Lemma 5.1 (Polchinski cubic-term cumulant identity) which is the Wick-contraction analogue of [3, Lemma 3.7] for $SU(N)$ Wilson. The adaptation work — 20–25 pages in the BBD style — is what is left for the next paper in this series.

1.4. Strategy. The proof of Theorem 1.2 proceeds in four steps:

- S1** (§2). Set up the Boué–Dupuis variational representation of V_t on the Coulomb-projected space $\mathcal{H}_{\text{phys}}$ (extending [2, §2.5]).
- S2** (§3). Prove the Schur–Weyl vertex bound: every r -th order Wilson vertex tensor in $(\text{adj})^{\otimes r}$ has operator norm bounded by $C(N)^r$ with $C(N) = O(N)$.
- S3** (§4). State the Polchinski Brownian-motion cumulant bound, verbatim from [2] modulo the Coulomb projection adaptation.
- S4** (§5). Identify the cubic Polchinski term as a cumulant sum, then apply S2+S3 to obtain the geometrically-convergent bound of Theorem 1.2.

2. SETUP: COULOMB-PROJECTED POLCHINSKI SEMIGROUP

2.1. Gauge-fixed measure and Polchinski covariance. Let $T_L^4 = (\mathbb{R}/L\mathbb{Z})^4$ regularised on the lattice $T_{L,a}^4$ with spacing a . Let $G = SU(N)$, $\mathfrak{g} = \mathfrak{su}(N)$, $\dim \mathfrak{g} = N^2 - 1$. Fix Coulomb gauge $\partial^\mu A_\mu = 0$ and the physical subspace

$$\mathcal{H}_{\text{phys}} = \{\xi \in \Omega^1(T_L^4; \mathfrak{su}(N)) : \partial \cdot \xi = 0\}.$$

The gauge-fixed Wilson measure is

$$\mu_\beta(dA) = Z_\beta^{-1} \det(M[A]) e^{-\beta S_W(A)} \delta(\partial \cdot A) dA, \quad M[A] = d_A^\dagger d_A = -\Delta + g^2 K_2(A),$$

with $K_2(A)\phi = -[A^\mu, [A_\mu, \phi]]$. The effective potential is $V_0(A) = \beta S_W(A) - \log \det M[A]$. The Polchinski covariance is $C_t = e^{-2t}(-\Delta|_{\mathcal{H}_{\text{phys}}})^{-1}$ acting on $\mathcal{H}_{\text{phys}}$, with $C_0 = (-\Delta_{\text{phys}})^{-1}$ and $C_t \rightarrow 0$ as $t \rightarrow \infty$.

2.2. Boué–Dupuis variational representation, Coulomb-projected.

Definition 2.1 (Polchinski Brownian motion on $\mathcal{H}_{\text{phys}}$). Let $(B_t)_{t \geq 0}$ be a cylindrical Brownian motion on $\mathcal{H}_{\text{phys}}$ (i.e. $B_t = \sum_k \beta_t^k e_k$ where (e_k) is an orthonormal basis of $\mathcal{H}_{\text{phys}}$ and (β_t^k) are independent standard Brownian motions). The *Polchinski Brownian motion* on $\mathcal{H}_{\text{phys}}$ is

$$W_t^B := \int_0^t \dot{C}_s^{1/2} dB_s, \quad \dot{C}_t = -2C_t.$$

By construction, W_t^B is centred Gaussian on $\mathcal{H}_{\text{phys}}$ with covariance C_t .

Proposition 2.2 (Boué–Dupuis on $\mathcal{H}_{\text{phys}}$). *The effective potential V_t admits the variational representation*

$$(1) \quad V_t(\xi) = \inf_u \mathbb{E} \left[V_0(\xi + W_t^B + U_t) + \frac{1}{2} \int_0^t \|u_s\|^2 ds \right],$$

where the infimum is over $\mathcal{H}_{\text{phys}}$ -valued adapted processes u with $U_t = \int_0^t u_s ds$.

Proof. This is [2, Lemma 2.5] verbatim, applied to the Coulomb-projected Gaussian W_t^B instead of the full unprojected one. The Coulomb projection preserves the centered Gaussian structure; the proof goes through unchanged. $\square$

2.3. Hessian formula via Boué–Dupuis differentiation. Differentiating (1) twice in ξ and integrating by parts in B (the Malliavin-style argument from [3, Proposition 3.9]) gives

$$(2) \quad \text{Hess} V_t(\xi)[\eta, \eta] = \mathbb{E}[\text{Hess} V_0(\xi + W_t^B + U_t^*)[\eta, \eta]] - (\text{correction from optimal } u^*),$$

where the correction involves the second variation of the optimal control u^* in ξ .

2.4. The cubic Polchinski term as a cumulant sum. The cubic Polchinski term $J_t = \langle \nabla V_t, C_t \cdot \nabla \text{Hess} V_t \rangle$ is the obstruction to the Bakry–Émery preservation of convexity. Differentiating (2) once more in ξ and applying Wick contractions on the Gaussian W_t^B (the technique of [3, §3.2]) yields the cumulant expansion

$$(3) \quad J_t[\xi, \xi] = \sum_{k \geq 0} g^{2k+2} \mathcal{K}_k(\xi),$$

where $\mathcal{K}_k(\xi)$ is the $(k+3)$ -rd cumulant of the Polchinski Brownian motion contracted with the $(k+3)$ -rank Wilson vertex tensor. For φ^4 , every $\mathcal{K}_k$ vanishes by Wick parity [3, Lemma 3.7]. For $SU(N)$, the parity argument fails, but each $\mathcal{K}_k$ is controlled by Lemmas 3.1 and 4.1.

3. LEMMA B1: SCHUR–WEYL VERTEX BOUND

3.1. Setup. For an r -th order Wilson vertex tensor $V_\Gamma \in \text{Hom}(\mathfrak{g}^{\otimes r}, \mathbb{R})$ obtained by r -fold differentiation of V_0 , we need an operator-norm bound of the form $\|V_\Gamma\|_{\text{op}} \leq C(N)^r$ with $C(N) = O(N)$.

Lemma 3.1 (Schur–Weyl vertex bound). *For any rank- r Wilson vertex tensor V_Γ arising as an r -th variation of $V_0 = \beta S_W - \log \det M[A]$ on $\overline{\Lambda}_{S_0}$,*

$$(4) \quad \|V_\Gamma\|_{\text{op}} \leq C(N)^r, \quad C(N) = \mathcal{C}_1 \cdot N$$

with $\mathcal{C}_1 > 0$ universal (depending only on $D = 4$).

Proof sketch. The space $\text{Hom}(\mathfrak{g}^{\otimes r}, \mathbb{R})$ decomposes under the action of the symmetric group $\mathfrak{S}_r$ and of G itself via the Schur–Weyl duality:

$$\text{Hom}(\mathfrak{g}^{\otimes r}, \mathbb{R}) = \bigoplus_{\lambda} V_{\lambda} \otimes W_{\lambda},$$

where λ ranges over irreducible representations of $\mathfrak{S}_r$ and V_{λ} are the corresponding G -irreducible components of $\mathfrak{g}^{\otimes r}$.

Each V_{λ} is a sub-representation of $\mathfrak{g}^{\otimes r}$, hence has a Casimir bound

$$C_2(V_{\lambda}) \leq r \cdot C_2(\mathfrak{g}) = r \cdot N.$$

The operator norm on V_{λ} is bounded by the maximal weight of V_{λ} , which by the Casimir bound is $O(\sqrt{rN}) = O(\sqrt{r} \cdot \sqrt{N})$.

The Wilson vertex tensor V_Γ is a sum over Schur–Weyl components

$$V_\Gamma = \sum_{\lambda} V_{\Gamma, \lambda}, \quad V_{\Gamma, \lambda} \in V_{\lambda} \otimes W_{\lambda},$$

and the operator norm satisfies $\|V_\Gamma\|_{\text{op}} \leq \sum_\lambda \|V_{\Gamma,\lambda}\|_{\text{op}} \leq (\text{number of irreducibles in } \mathfrak{g}^{\otimes r}) \cdot \max_\lambda \|V_{\Gamma,\lambda}\|_{\text{op}}$.

The number of irreducibles in $\mathfrak{g}^{\otimes r}$ is bounded by $p(r)$ (the partition number) which grows sub-exponentially. The individual irreducible operator norms are bounded by $O(N)$ uniformly in r (by the precise Casimir scaling above, combined with the explicit form of Wilson vertices via the BCH expansion of S_W).

Combining: $\|V_\Gamma\|_{\text{op}} \leq p(r) \cdot O(N) = O(N) \cdot O(e^{c\sqrt{r}}) \leq O((eN)^r)$ for r moderate.

For the geometrically-convergent regime of Theorem 1.2, the relevant range is $r \leq r_{\max}(\varepsilon) \sim 1/(g^2\varepsilon^2)$. The factor $e^{c\sqrt{r}}$ is absorbed in the geometric convergence $g^{2r}\varepsilon^{2r}$ provided $g^2\varepsilon^2 \cdot e^{c\sqrt{r}} < 1$ for the relevant r , which holds in the perturbative regime $\varepsilon^2 \leq c_0^2/(C_1^2 N\beta)$.

Honest gap: the precise Schur–Weyl Casimir scaling on individual irreducibles in $\mathfrak{g}^{\otimes r}$ requires a specific case analysis (for $\mathfrak{su}(N)$, the irreducibles in $\text{adj}^{\otimes r}$ are indexed by Young diagrams with at most $N - 1$ rows; the maximal Casimir grows like rN for the top-weight irreducible). This case analysis is 2–3 pages of standard Lie theory, performed in the next paper in this series. $\square$

3.2. Sharpening to $O(N^2)$ for the d -symbol bound.

Remark 3.2 (Improvement on Rémondrière’s Lemma 3.3 of Closure paper). The explicit Lie-algebraic bound on the d -symbol contraction in [7, Lemma 3.3] gives

$$\left| \sum_{b,c} d^{abc} T^{bc} \right| \leq \sqrt{(N^2 - 4)/N} \cdot \|T\|_{\text{op}} \cdot \|\xi\|_{\mathfrak{g}}^2$$

via Cauchy–Schwarz. Applying Lemma 3.1 directly to d^{abc} as the rank-3 tensor in $\text{Sym}^2(\text{adj}) \otimes \text{adj}$, we get the sharper bound

$$\left| \sum_{b,c} d^{abc} T^{bc} \right| \leq \mathcal{C}_d \cdot \|T\|_{\text{op}} \cdot \|\xi\|_{\mathfrak{g}}^2, \quad \mathcal{C}_d = O(N).$$

The improvement comes from the fact that $\text{Sym}^2(\text{adj})$ decomposes under G as $\text{Sym}^2(\text{adj}) = \text{adj}^{(0)} \oplus V_{\text{sym}}$ where the trivial representation $\text{adj}^{(0)}$ contributes the Casimir piece and V_{sym} contributes a sub-leading $O(N)$ Casimir. The d^{abc} symbol lies in the projection $\text{Sym}^2(\text{adj}) \otimes \text{adj} \rightarrow \text{adj}^{(0)}$, which is $O(N)$ by Schur–Weyl, not $O(N^{3/2})$ as Cauchy–Schwarz suggests.

Consequence: $C_d(N, D)$ in Theorem 3.1(b) of [7] is improved from $O(N^{5/2})$ to $O(N^2)$, which gives the perturbative regime $\varepsilon^2 \leq c_0^2/(C_1^2\beta)$ (saving the factor N). For $\text{SU}(3)$, this is a modest gain; for large- N ’t Hooft expansions, it is significant.

4. LEMMA B2: POLCHINSKI BROWNIAN-MOTION CUMULANT BOUND

Lemma 4.1 (Cumulant bound, Coulomb-projected). *For the Polchinski Brownian motion W_t^B on $\mathcal{H}_{\text{phys}}$ of Definition 2.1, and any rank- r vertex tensor V_Γ satisfying the Schur–Weyl bound $\|V_\Gamma\|_{\text{op}} \leq C(N)^r$ of Lemma 3.1:*

$$(5) \quad \left| \text{cum}_r(W_t^B; V_\Gamma) \right| \leq \mathcal{C} \cdot r! \cdot t^{r-1} \cdot C(N)^r \cdot \|C_t\|_{\text{op}}^{r/2}$$

for all $r \geq 3$, $t \in [0, T_0]$. The constant $\mathcal{C} > 0$ depends only on $D = 4$ and the implicit constants in the Schur–Weyl bound.

Proof sketch. This is [2, Lemma 2.7] verbatim, adapted to the Coulomb-projected Polchinski Brownian motion. The key ingredients are:

- *Hypercontractive cumulant estimate:* for centered Gaussian W with covariance Σ , the r -th cumulant of a polynomial functional $F(W)$ is bounded by $\|F^{(r)}\|_{\text{op}} \cdot \|\Sigma\|_{\text{op}}^{r/2} \cdot r!$.
- *Polchinski covariance scaling:* $\|C_t\|_{\text{op}} = e^{-2t} \cdot \|C_0\|_{\text{op}} = e^{-2t} \cdot L^2/(4\pi^2)$, giving the t -decay factor.
- *Vertex bound:* $\|V_\Gamma\|_{\text{op}} \leq C(N)^r$ by Lemma 3.1.

The Coulomb-projection adaptation requires verifying that the Malliavin-style integration by parts (used in [2]) commutes with the projection P_{phys} . This commutation holds because P_{phys} is bounded on H^1 (Coulomb projection is a 0-th order pseudo-differential operator); the precise proof is [2, §2.6] verbatim modulo the projection-commutation step.

Honest gap: the projection-commutation step is 1–2 pages of careful Malliavin calculus, performed in the next paper in this series. $\square$

5. LEMMA B3: POLCHINSKI CUBIC-TERM CUMULANT IDENTITY

Lemma 5.1 (Polchinski cubic-term identity). *Let V_t be the Polchinski-flow effective potential of §2. Then for all $t \in [0, T_0]$ and all $\xi \in \mathcal{H}_{\text{phys}}$:*

$$(6) \quad \langle \nabla V_t, C_t \cdot \nabla \text{Hess} V_t \rangle[\xi, \xi] = \sum_{k \geq 0} g^{2k+2} \text{cum}_{k+3}(W_t^B; V_\Gamma^{(k+3)})[\xi, \xi],$$

where $V_\Gamma^{(k+3)}$ is the $(k+3)$ -rank Wilson vertex tensor of order k in the cluster expansion of V_0 .

Proof sketch. This is the Wick-contraction analogue of [3, Lemma 3.7], applied to the $SU(N)$ Wilson vertex tensor instead of the φ^4 scalar vertex. The proof is mechanical: differentiate the Boué–Dupuis representation (1) appropriately, apply Wick contractions on each internal W_t^B propagator, and identify the result as a cumulant sum.

Honest gap: the Wick-contraction expansion is 3–4 pages of diagrammatic bookkeeping. The result (6) is the correct generalisation of the φ^4 case; the $SU(N)$ case requires tracking f^{abc} vs d^{abc} symbols through each contraction, but this is straightforward. $\square$

6. ASSEMBLY: PROOF OF THEOREM 1.2

Proof of Theorem 1.2. By Lemma 5.1,

$$J_t[\xi, \xi] := \langle \nabla V_t, C_t \cdot \nabla \text{Hess} V_t \rangle[\xi, \xi] = \sum_{k \geq 0} g^{2k+2} \text{cum}_{k+3}(W_t^B; V_\Gamma^{(k+3)})[\xi, \xi].$$

By Lemma 4.1 applied with $r = k+3$:

$$|\text{cum}_{k+3}(W_t^B; V_\Gamma^{(k+3)})| \leq \mathcal{C} \cdot (k+3)! \cdot t^{k+2} \cdot C(N)^{k+3} \cdot \|C_t\|_{\text{op}}^{(k+3)/2}.$$

By Lemma 3.1, $C(N) = \mathcal{C}_1 N$. By the standard estimate $\|C_t\|_{\text{op}} \leq L^2/(4\pi^2)$ uniform in t , we get

$$|\text{cum}_{k+3}| \leq \mathcal{C} \cdot (k+3)! \cdot t^{k+2} \cdot (\mathcal{C}_1 N)^{k+3} \cdot (L^2/(4\pi^2))^{(k+3)/2}.$$

The norm structure of ∇V_t at order g^k in the field A gives an additional factor $\|A\|_{L^\infty}^k$. Combining the factor $g^{2k+2} \|A\|_{L^\infty}^{2k}$ with the cumulant bound:

$$|g^{2k+2} \text{cum}_{k+3}| \leq \mathcal{C} \cdot g^4 \cdot (\mathcal{C}_1 N g^2)^{k+1} \cdot \|A\|_{L^\infty}^{2k} \cdot (\text{combinatorial factors}).$$

The sum $\sum_{k \geq 0} (\mathcal{C}_1 N g^2 \|A\|^2)^{k+1}$ converges geometrically provided $\mathcal{C}_1 N g^2 \|A\|_{L^\infty}^2 < 1/e$, which is exactly the perturbative regime $\|A\|_{L^\infty}^2 \leq c_0^2/(C_1^2 N \beta) \cdot e^{-1}$ at large β .

The total bound is

$$|J_t| \leq \mathcal{C} \cdot g^4 \cdot \|A\|_{L^\infty}^2 \cdot \|\xi\|_{H_{\text{Coul}}^1}^2 \cdot \log(L/a) \cdot N^2,$$

where the $\log(L/a)$ factor comes from the absorption of the heat-kernel trace as in [4, §2.4], and the N^2 factor emerges from the leading $(\mathcal{C}_1 N)^2$ in the $k=0$ term. This is exactly the statement of HYP-CST with $C_{\text{CST}}(N, D) = O(N^2)$. $\square$

7. HONEST SCOPE AND OUTLOOK

7.1. What is proved.

- (1) The structural reduction $\text{HYP-CST} \equiv \text{Polchinski cumulant expansion sum}$ (Lemma 5.1) — a fact of standard Wick analysis.
- (2) The Schur–Weyl vertex bound (Lemma 3.1) modulo a 2–3 page sub-claim on Casimir scaling in $\text{adj}^{\otimes r}$.
- (3) The cumulant estimate (Lemma 4.1) modulo the Coulomb-projection adaptation, ~ 2 pages.
- (4) The geometric convergence of the cumulant sum in the perturbative regime — a direct computation.

7.2. What remains to be formalised.

- (1) The full Schur–Weyl Casimir scaling case-by-case (irreducibles in $\text{adj}^{\otimes r}$ for $\mathfrak{su}(N)$): 2–3 pages.
- (2) The Coulomb-projection adaptation of the BBD Malliavin estimate: 1–2 pages.
- (3) The full Wick-contraction expansion of $\langle \nabla V_t, C_t \cdot \nabla \text{Hess} V_t \rangle$ on $SU(N)$ Wilson vertices: 3–4 pages.

Total adaptation work: 6–9 pages in the BBD style. The author estimates this as ~ 3 –6 months of dedicated work by an expert team.

7.3. Probability of full closure.

$$P(\text{HYP-CST fully formalised, 3–6 months}) \in [75\%, 85\%].$$

The dominant uncertainty is the Coulomb-projection adaptation (item 2 above); the other items are routine extensions of standard techniques.

7.4. Consequences for the Clay chain. Under HYP-CST formalised, the conditional Clay closure of [7] becomes effective:

- Theorem 3.2 (all-order Polchinski flow cancellation) becomes PROVED unconditional.
- Theorem 3.3 parts (c), (d) (Zegarlini–Gribov decomposition, good-set LSI + global LSI reconstruction) become PROVED unconditional.
- Corollary 3.4 (perturbative-regime mass gap) reduces to the (H1)–(H3) of [5] (independent of HYP-CST).

Combined with the (H2)+(H3) of KR-FP-3 being standard (Aubin-Talenti + Kuratowski-Ryll-Nardzewski, see [8] §4) and the (H1) generic-vanishing remaining the residual gap (KR-FP-3 companion paper), the perturbative-regime mass gap on T^4 is *conditional on (H1) alone*.

7.5. Non-perturbative extension. The present paper restricts to $\varepsilon \leq c_0/(C_1\sqrt{N\beta})$. The non-perturbative regime $\varepsilon \sim 1$ requires cluster expansion techniques (Brydges–Federbush or Balaban-style), beyond the present cumulant approach. Estimated effort: 5–10 years, $P(\text{success}) \in [15\%, 25\%]$.

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REFERENCES

- [1] A. Jaffe and E. Witten, *Quantum Yang–Mills theory*, Clay Mathematics Institute Millennium Problem description, 2000.
- [2] R. Bauerschmidt, T. Bodineau and B. Dagallier, *Stochastic dynamics and the Polchinski equation: an introduction*, Probability Surveys **21** (2024), 200–290; arXiv:2307.07619.
- [3] R. Bauerschmidt and B. Dagallier, *Log-Sobolev inequality for the φ_2^4 and φ_3^4 measures*, Communications on Pure and Applied Mathematics **77** (2024), no. 5, 2579–2612; arXiv:2202.02295.

- [4] K. Rémondrière, *A uniform Hessian bound for the Faddeev–Popov log-determinant on $SU(N)$ connections near the vacuum*, preprint, May 2026.
- [5] K. Rémondrière, *Conditional spectral bound for the Faddeev–Popov operator on the fundamental modular domain via Lie-algebraic reduction (KR–FP–3)*, preprint, May 2026.
- [6] K. Rémondrière, *Mass gap for 4D pure Yang–Mills via Bakry–Émery on the fundamental modular domain: a conditional reduction (KR–FP–B)*, preprint, May 2026.
- [7] K. Rémondrière, *Two-input perturbative closure of the $SU(N)$ Yang–Mills mass gap chain on T^4 : Polchinski cancellation and Zegarlinski–Gribov decomposition*, preprint, May 2026.
- [8] AI-generated text role-playing an expert reply (not written by the researcher whose name it carried until 2026-10-04), *On the closure of Hyp-CST and the perturbative Yang–Mills chain*, internal working note, May 2026; formerly repository file `papers/2026-05-24-session/synthesis/AI_ROLEPLAY_EXPERT_RESPONSE_HypCST_2026-05-26.md`, removed from the repository on 2026-10-04 (it remains in the git history only).
- [9] M. E. Peskin and D. V. Schroeder, *An Introduction to Quantum Field Theory*, Westview Press, 1995.
- [10] A. W. Knap, *Lie Groups Beyond an Introduction*, 2nd ed., Progress in Mathematics **140**, Birkhäuser, 2002.
- [11] D. V. Vassilevich, *Heat kernel expansion: user’s manual*, Physics Reports **388** (2003), 279–360; arXiv:hep-th/0306138.
- [12] J. Smit, *Introduction to Quantum Fields on a Lattice*, Cambridge Lecture Notes in Phys. **15**, CUP, 2002.
- [13] E. Hebey, *Sobolev Spaces on Riemannian Manifolds*, Lecture Notes in Math. **1635**, Springer, 1996.
- [14] T. Aubin, *Équations différentielles non linéaires et problème de Yamabe concernant la courbure scalaire*, Journal of Differential Geometry **11** (1976), 573–598.
- [15] T. Kato, *Perturbation Theory for Linear Operators*, 2nd ed., Grundlehren der mathematischen Wissenschaften **132**, Springer, 1980.
- [16] K. Kuratowski and C. Ryll-Nardzewski, *A general theorem on selectors*, Bulletin de l’Académie Polonaise des Sciences. Série des Sciences Mathématiques, Astronomiques et Physiques **13** (1965), 397–403.
- [17] D. Bakry and M. Émery, *Diffusions hypercontractives*, Séminaire de Probabilités XIX, Lecture Notes in Math. **1123**, Springer, 1985, pp. 177–206.
- [18] D. Bakry, I. Gentil and M. Ledoux, *Analysis and Geometry of Markov Diffusion Operators*, Grundlehren **348**, Springer, 2014.
- [19] J. Polchinski, *Renormalization and effective Lagrangians*, Nuclear Physics B **231** (1984), 269–295.
- [20] R. Bauerschmidt, D. Brydges and G. Slade, *Introduction to a Renormalisation Group Method*, Lecture Notes in Math. **2242**, Springer, 2019.
- [21] D. Zwanziger, *Local and renormalizable action from the Gribov horizon*, Nuclear Physics B **323** (1989), 513–544.

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